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#include <stdio.h>
#include <math.h>

int main() {
    // Location Coordinates (X, Y)
    int gateX = 0, gateY = 0;
    int libraryX = 4, libraryY = 3;
    int labX = 8, labY = 6;

    int startX, startY, endX, endY;
    int choice, transport, weather;
    double distance, speed, time;

    printf("=====\n");
    printf("        SMART CAMPUS GUIDE NAVIGATOR        \n");
    printf("=====\n");
    printf("Available Locations:\n");
    printf("1. Main Gate      (0, 0)\n");
    printf("2. Library        (4, 3)\n");
    printf("3. Computer Lab   (8, 6)\n");
    printf("-----\n");

    // Select Starting Point
    printf("Enter your CURRENT location (1-3): ");
    if (scanf("%d", &choice) != 1 || choice < 1 || choice > 3) {
        printf("Error: Invalid selection. Program exiting.\n");
        return 1;
    }

    if (choice == 1) { startX = gateX; startY = gateY; }
    else if (choice == 2) { startX = libraryX; startY = libraryY; }
    else { startX = labX; startY = labY; }

    // Select Destination Point
    printf("Enter your DESTINATION (1-3): ");
    if (scanf("%d", &choice) != 1 || choice < 1 || choice > 3) {
        printf("Error: Invalid selection. Program exiting.\n");
        return 1;
    }

    if (choice == 1) { endX = gateX; endY = gateY; }
    else if (choice == 2) { endX = libraryX; endY = libraryY; }
    else { endX = labX; endY = labY; }

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// Select Mode of Transport
printf("\nChoose Mode of Transport:\n1. Walking (5 km/h)\n2. Biking (15 km/h)
if (scanf("%d", &transport) != 1 || (transport != 1 && transport != 2)) {
    printf("Error: Invalid transport mode. Program exiting.\n");
    return 1;
}

// Select Weather Condition
printf("Is it raining? (1 for Yes / 0 for No): ");
if (scanf("%d", &weather) != 1 || (weather != 0 && weather != 1)) {
    printf("Error: Invalid weather option. Program exiting.\n");
    return 1;
}

// Calculate Distance using Distance Formula:  $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$ 
distance = sqrt(pow(endX - startX, 2) + pow(endY - startY, 2));

// Determine Base Speed
if (transport == 1) {
    speed = 5.0; // Walking
} else {
    speed = 15.0; // Biking
}

// Adjust Speed for Rain (30% reduction)
if (weather == 1) {
    speed = speed * 0.7;
}

// Calculate Estimated Travel Time in Minutes
time = (distance / speed) * 60;

// Display Results
printf("\n===== \n");
printf("                NAVIGATION ROUTE                \n");
printf("===== \n");
printf("Total Distance : %.2f km\n", distance);
printf("Estimated Time : %.1f minutes\n", time);
printf("----- \n");
printf("Smart Recommendation: ");

if (weather == 1 && transport == 2) {
    printf("It is raining! Biking can be slippery. Please ride slow and wear
} else if (weather == 1) {
    printf("It is raining! Consider taking covered walkways or carrying an um
} else if (distance == 0) {
    printf("You are already at your destination!\n");

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    } else {  
        printf("Weather conditions are ideal. Have a safe journey!\n");  
    }  
    printf("=====\n");  
  
    return 0;  
}
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